

System Requirements

Hardware Requirements

- Development Machine:
 - CPU: Multi-core (i5/i7 or equivalent)
 - RAM: 16GB or more (especially if you're handling large files for RAG)
 - GPU: Optional for model training, but helpful for faster inference
- Server :
 - Specs:
 - RAM: 16GB+
 - Storage: SSD for fast read/write

Software Requirements

Frontend/ Backend

A modern frontend framework like React(vite) paired with a CSS framework like Tailwind CSS is recommended. For quick UI development, consider utilizing shadcn/ui, a collection of pre-built components.

Node.js/Express or Python (Flask/FastAPI) are commonly used backend technologies. Choose the one that best aligns with your existing expertise and preferences.

Front/Back-end components:

1. Search Chat History
2. New Chat
3. Close Sidebar
4. Chat History Sidebar
5. Profiling
6. Chat Title
7. Chat responses
8. Upload File
9. Type Query box

Database

PostgreSQL or MongoDB are popular choices for storing user inputs, file metadata, and other application data.

Model Hosting

For accessing LLMs, LangChain and LLaMA powerful tools that streamline RAG (Retrieval-Augmented Generation) workflows. LangChain simplifies the integration of language models with external data sources, while LLaMA offers efficient, high-performance, open-weight models suitable for on-premise deployment.

File Handling

For efficient retrieval of files implement local storage during development. Ensure the storage solution is secure and scalable to meet your project's needs.

1. TextLoader (LangChain)
2. CSVLoader (LangChain)
3. UnstructuredExcelLoader
4. UnstructuredPDFLoader
5. UnstructuredWordDocumentLoader
6. UnstructuredPowerPointLoader